# Kappa Foundation

Investment committee memo | Revised trustee mandate | 28 October 2026

Decision as at 31 December 2022. Historical evaluation is reported separately.

## 1. Recommendation

We recommend funding the revised commitments with principal-matched bonds and keeping the equity weights selected for the original 22 October proposal. Our bond holdings are 2,400 Beta, 1,200 Gamma and 2,400 Delta units. Together they set aside the principal needed for the USD 2.4 million payment in 2024, USD 1.2 million in 2025 and USD 2.4 million in 2027. This uses USD 5,784,228 at the case-date quotes. After the stated 10 basis point equity entry fee, the remaining USD 6,209,562 is invested in the original Sample Mean-Variance equity portfolio.

We chose this route because the foundation cannot miss a scheduled learning-centre payment. A feasible alternative funds the 2025 liability partly with coupons and leaves about USD 354,098 more for shares. However, it has almost no cash left immediately after the 2025 payment. Principal matching instead retains approximately USD 396,000. That reserve matters if the foundation needs to respond to market losses or an unexpected payment. Our preference is a judgement about liquidity and reliability, not a claim that principal matching maximizes expected terminal wealth.

The 2030 target remains USD 8 million of 2022 purchasing power. At our 2.3% inflation assumption, that means approximately USD 9.596 million in 2030. Our recommended revised portfolio needs an equity CAGR of about 4.80% over eight years, assuming no return on the bond coupons held in cash. This is a target hurdle, not a forecast or guarantee.

Our main reservation is the equity concentration. The selected holdings are eight individual stocks, with AAPL and MSFT together at the 30% Technology sector cap. They earned positive returns in the later 2023 to 2025 test, but underperformed VTI. The investment committee should be given both sides of that evidence.

## 2. The changed payments and bond funding

The original case called for USD 2.4 million at the end of 2024 and USD 3.6 million at the end of 2027. The 23 October trustee update moves USD 1.2 million of the latter amount into 2025. The total payment commitment is still USD 6 million. Our valuation date remains 31 December 2022, so we use the bond prices and information available for that case date rather than later prices.

| Policy | Bond cost, USD | Net equities, USD | Cash retained 2030, USD | Equity CAGR hurdle |
|---|---|---|---|---|
| Original Beta and Delta | 5,764,524 | 6,229,247 | 630,000 | 4.66% |
| Revised principal matching | 5,784,228 | 6,209,562 | 564,000 | 4.80% |
| Revised coupon funding | 5,429,776 | 6,563,660 | 168,000 | 4.63% |

The original portfolio buys 2,400 Beta and 3,600 Delta units. Our proposed revision replaces 1,200 of the Delta units with 1,200 Gamma units, while keeping 2,400 Beta units. The bond cost rises by USD 19,704; the growth sleeve falls by USD 19,684 after the entry-fee convention. Those small changes have a disproportionate effect on the timing of available cash.

We recomputed the contractual cash flows from unit face values and coupons. Under principal matching the retained cash balances after the mandatory payments are USD 264,000 at the end of 2024, USD 396,000 at the end of 2025 and USD 564,000 at the end of 2027. The 2025 coupon-funded alternative has effectively no such cushion after the payment. Our analysis assumes any coupon received on a payment date is available for that payment, and unspent coupons earn zero thereafter.

Bond duration, convexity, spot-rate construction and parallel-rate as well as curve-twist scenarios are supplied in the original bond workstream. Illustrative +100 basis point bond repricing gives approximately a 3.08% immediate sleeve loss under principal matching. This is market-value exposure and does not imply that the fixed face-value payment disappears if bonds are held to maturity.

## 3. Choosing and sizing the equity sleeve

We used ten eligible US common-stock candidates and two US equity ETFs across six stock sectors. For the original case decision, the data consist of 96 monthly observations from January 2015 to December 2022. Forecast means blend 25% of each stock's sample mean with a 75% VTI-based prior. The weights are long-only, with a 15% cap for each direct stock, a 30% cap for direct holdings in the same sector, a 30% limit for each ETF and a 40% combined ETF limit.

The frozen allocation holds eight stocks and no ETFs. AAPL, MSFT and PG are each at 15%, while the Technology sector reaches 30%. The remaining positions are JPM, JNJ, PFE, KO and CAT. This is not as diversified as owning VTI, and the estimated optimum depends on assumptions about expected returns. We therefore tested three different portfolio methods rather than treating the first optimizer result as decisive.

Sample Mean-Variance is the reference portfolio at risk aversion A = 3. We compared it with a Single-Index covariance approach and a constrained Minimum Variance portfolio, then tested A = 6 and lower confidence in historical expected returns. The original Sample MV has estimated annual arithmetic return 12.05% and volatility 14.98%. Single-Index MV yields 11.91% and 14.46%; Minimum Variance yields 10.86% and 13.15%. All three risk figures use the same sample-covariance assessment, even where the optimizer uses the Single-Index assumption. These numbers are predictions from the historical estimation model, not results earned in later markets.

The revised principal bond budget changes the scale of investment, but not the normalized stock weights. Reoptimizing the sample model under the revised budget raises the estimated equity contribution to foundation utility only from 0.053328705 to 0.053328718 at A = 3. We do not think such a small modeled gain warrants changing the original stocks. The factor exposures of the frozen normalized sleeve remain unchanged across funding choices; its dollar risk contribution to the foundation does change.

## 4. Factors and forecast risk

We estimated a CAPM regression and a Fama-French three-factor regression for the frozen equity returns using the same 96 months and the archived Kenneth French market, size, value and risk-free series. Risk-free returns are subtracted once to create monthly excess returns. These factor models help explain the selected portfolio's historical co-movements; they are not a separate forecast of guaranteed performance.

| Historical regression measure | CAPM | FF3 |
|---|---|---|
| Market beta | 0.805 | 0.867 |
| Adjusted R squared | 0.785 | 0.844 |
| Monthly model alpha | 0.646% | 0.569% |
| SMB loading | Not included | -0.426 |
| HML loading | Not included | 0.010 |

The negative SMB estimate suggests a large-company tilt and the market beta below one suggests slightly lower market sensitivity than the broad market in this sample. The near-zero HML loading is consistent with the mixture of growth and value-style holdings. FF3 explains the sample better than CAPM; adding SMB and HML is jointly significant under the stated OLS test.

The alpha is positive and statistically significant in sample. However, the candidates were chosen with hindsight, the same estimation window helped select weights, and the model omits other possible sources of return. We cannot attribute the intercept to investment skill or assume it will persist. For perspective, MSFT has higher market beta than PG and PG's individual FF3 alpha is not statistically distinguishable from zero.

The revised funding date changes the bond allocation rather than the chosen equity return series. A new factor table checks 15 combinations of three funding cases and five stock policies. The three scenarios with the frozen normalized weights are identical at the monthly-return level, so their CAPM and FF3 estimates also agree. A portfolio with reoptimized stock weights does need its own regression; those estimates are saved in the factor results.

## 5. Sensitivities, downside risk and a change of mind

Our decisions depend on uncertain growth, inflation and market losses. In the base case, the revised principal-matching portfolio needs about 4.80% compound annual equity growth to reach the 2030 purchasing-power goal. A 36-case table varies inflation and the starting operating cash buffer. A cash buffer means less invested in shares; we keep it unspent at a zero return in those cases so it is not double-counted as both available for spending and part of 2030 wealth.

In a simultaneous illustrative equity fall of 25% and 100 basis point increase in bond yields, the starting fund loses about 14.42% under principal matching and about 15.08% under coupon funding. We propose a 12% early warning and a 15% escalation based on rolling, withdrawal-adjusted fund returns. The trustees have not approved these precise thresholds or the measurement period; they are our response to the board's stated discomfort with losses near 15%, not established trustee rules. It would be incorrect to state that the simulated 15.08% figure proves a rule was breached.

For the same two fixed portfolios, the modeled preference switches from frozen Sample MV to Minimum Variance at risk aversion A near 8.94 under principal funding and 8.46 under coupon funding. Lower trust in historical means can produce a similar change, with the revised principal reference near confidence weight 0.0839. These are pairwise fixed-portfolio comparisons. An investor who changes A should consider reoptimizing under the new preference rather than automatically purchasing the old Minimum Variance weights.

We would ask the trustees to reconsider the equity approach if their true loss tolerance is lower than assumed, forecast confidence deteriorates, or expected inflation is persistently higher than the 2.3% base. We would also reconsider holding a larger cash reserve if the timetable or certainty of payments changed. These conditions are more defensible than a promise that one optimizer will remain optimal until 2030.

## 6. The frozen 2023 to 2025 market test

After the original policy was fixed, the team obtained the daily Yahoo price series covering the later evaluation window and saved the raw responses, versions and hashes. The eight-stock portfolio and a VTI-only alternative were both charged the same initial 10 basis point entry fee. This evaluation starts from the original policy's USD 6.229 million growth allocation, not the revised principal-matching allocation of USD 6.210 million shown elsewhere. The observed equity valuation uses the 3 January 2023 closing-price proxy, keeps share quantities fixed and permits the weights to drift. It covers 752 common trading dates and 36 month-end observations.

| Growth-sleeve statistic | Selected stocks | VTI |
|---|---|---|
| CAGR after entry fee | 17.69% | 22.45% |
| Annualized monthly volatility | 11.80% | 12.60% |
| Largest monthly NAV drawdown | -8.70% | -9.11% |
| End-2025 equity value, USD million | 10.15 | 11.43 |

The arithmetic annualized active return was -4.11% and annualized tracking error was 6.91%. The portfolio earned money but lost ground against the simpler broad-market alternative. That weakens the case for claiming we have a reliable advantage in stock selection. It does not allow us to use 2023 to 2025 returns to remake the 2022 decision.

JNJ offered shareholders an optional exchange into Kenvue in 2023. The original no-voluntary-exchange policy therefore keeps JNJ and receives no automatic KVUE shares. A separate no-tender calculation using JNJ close prices and ordinary cash dividends differs from the Yahoo adjusted-price path by about USD 188 at the entire equity portfolio level by December 2025. This supports our treatment, with a remaining caveat about actual dividend-reinvestment timing and vendor adjustments.

This evaluation is for the traded equity sleeve only. The case bonds have no observed 2023 to 2025 trading-price history. A separate modeled whole-foundation monitor uses the case YTMs, the revised liability schedule and observed equity prices; it must not be presented as observed total-fund performance. An actual university email was sent on 22 October at 15:44:53 Vietnam time with the original ZIP as a first draft; its message asks whether Moodle submission is also needed. This documents transmission, not LMS receipt or instructor acceptance, and the exact emailed ZIP-member SHA256s remain unverified. We therefore report the price-based results as conditional on final original-submission verification.

## 7. Implementation and review

On the case date we allocate USD 5.784 million to the principal-matched bonds and invest the remaining USD 6.210 million net in the original Sample MV weights. We do not voluntarily exchange JNJ or rebalance the original 2023 to 2025 evaluation portfolio. We keep the bond coupons received after payment dates as cash without assuming a return until the 2030 goal date. The investment committee receives a regular account of bond cash needs, equity values, any cash shortfalls and the 12-month withdrawal-adjusted return.

The fund monitor distinguishes between observed stock prices and modeled bond valuations. It adds back a mandatory distribution in the month of payment when computing a time-weighted return. Under our proposed monitoring policy, a 12% rolling loss would prompt an early review of liquidity and forecasts, while a 15% rolling loss would trigger a deeper trustee discussion, subject to board approval. Neither action changes the payment obligations or licenses use of future price information in a 2022 decision.

Several limits remain. Our initial expected-return estimates can be unstable; all eight chosen stocks are US-listed and none of the candidate ETFs was selected. The SEC case-date register contains traceable URLs but not all original filing bytes. French and Yahoo series are saved from retrievals performed in October 2026 and may incorporate retrospective adjustments. The growth evaluation has actual archived prices, but the course receipt cannot yet be independently authenticated. Finally, no author may claim that a numerical quality check is the same as a signed independent review.

We would keep principal matching and the frozen equity weights for the committee recommendation, while making the coupon alternative, Minimum Variance comparison and VTI underperformance visible. We have not assumed successful funding of the 2030 purchasing-power goal. The team must make the final submission from one integrated set of numbers and source files, disclose material AI assistance and confirm that the decision and reviewer records reflect actual contributions.

## References and file guide

[1] BFIN WS2026 Group Assignment, original four-page case and 23 October 2026 trustee update provided to the group.

[2] Kappa initial decision record and preserved 2022 policy evidence. The exact emailed 22 October ZIP has not been independently authenticated in this repository.

[3] Bond model: bond_funding/Kappa_Bond_Funding_Model_Revised.xlsx and bond_funding/outputs/revised.

[4] Equity model: portfolio_construction/Portfolio Final/Results.xlsx and portfolio_construction/revised_mandate.

[5] Factor model: factor_analysis/outputs/factor_coefficients.csv and revised_mandate_factor_scenarios.csv. French factor data are archived in factor_analysis/data/raw/french.

[6] Market evaluation: portfolio_construction/Portfolio Final/Sources, including Yahoo Raw.zip and Market Source.json, with dates and checksums.

[7] Johnson & Johnson. Voluntary exchange offer announcement, 24 July 2023; final results, 23 August 2023. https://www.jnj.com/media-center

[8] Quality checks and decisions: quality_assurance/qa_report.json and quality_assurance/decision_log.csv.

The students report taking responsibility for the investment idea, methods and lecture formulas, parameters, securities and sources, feasibility checks, interpretation and presentation decisions. AI materially supported code, source/data work, calculations, testing, layout implementation and editing under their direction. All results and limitations remain reviewable. The group reports collective agreement, but individual signatures and LMS/instructor acceptance have not been independently authenticated.